

Project Report

Omnichannel Retail Sales & Inventory Analytics

1. Executive Summary

This project focuses on building an end-to-end retail analytics solution for an omnichannel retail business. The objective was to transform raw retail sales and inventory data into meaningful business insights through data cleaning, exploratory analysis, SQL-based analytics, and interactive Power BI dashboards.

The project enables business stakeholders to monitor sales performance, inventory movement, customer behavior, regional trends, and procurement requirements. By combining SQL Server, Excel, Python, Power BI, Git, and GitHub, the team developed a complete analytics workflow that supports data-driven decision-making.

2. Business Problem

Retail businesses often struggle to consolidate information from multiple sales channels, stores, products, and inventory systems. Without proper analytics, decision-makers face challenges such as:

- Limited visibility into sales performance.
- Difficulty identifying top-performing products and regions.
- Poor inventory planning leading to stockouts or dead stock.
- Lack of understanding of seasonal demand patterns.
- Inability to track growth trends across locations.

This project addresses these challenges through centralized analytics and stakeholder-specific dashboards.

3. Project Objectives

The primary objectives of this project were:

- Analyze retail sales performance across products, stores, cities, and regions.
- Monitor inventory movement and identify dead stock.
- Perform trend analysis using monthly and yearly growth metrics.
- Build stakeholder-focused Power BI dashboards.

- Generate actionable business recommendations.
 - Implement a collaborative GitHub workflow for project management.
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4. Technology Stack

Data Storage & Analysis

- SQL Server

Data Cleaning

- Microsoft Excel

Exploratory Data Analysis

- Python

Data Visualization

- Power BI

Version Control

- Git
 - GitHub
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5. Dataset Overview

The project uses a retail dataset for a real-world omnichannel retail operations.

Fact Tables

fact_orders

Contains transactional sales data.

fact_inventory

Contains inventory and stock movement data.

Dimension Tables

dim_product

Product information and categories.

dim_customer

Customer information.

dim_stores

Store, city, and regional information.

dim_date

Date hierarchy used for trend analysis.

6. Project Architecture

Raw Dataset

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Excel Cleaning & Validation

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Python Exploratory Data Analysis

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SQL Server Database

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Business Analysis & SQL Views

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Power BI Data Model

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Interactive Dashboards

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Business Insights & Report

7. Data Cleaning & Validation

The dataset was cleaned and validated before loading into SQL Server.

Key activities included:

- Handling missing values.
- Standardizing date formats.

- Validating data types.
 - Identifying duplicate records.
 - Correcting inconsistent values.
 - Verifying sales amount calculations.
 - Validating product and customer mappings.
 - Creating additional columns as per the requirements.
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8. SQL Analysis

Sales Performance Analysis

Performed KPI calculations including:

- Total Revenue
- Total Orders
- Total Quantity Sold
- Average Order Value (AOV)

Additional analysis included:

- Product performance analysis
 - Category performance analysis
 - Customer purchasing behavior
 - Store performance analysis
 - Regional sales analysis
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Inventory Analysis

The inventory analysis focused on:

- Fast-moving products
- Mid-moving products
- Slow-moving products
- Dead stock identification
- Stockout risk analysis
- Procurement planning

A procurement recommendation framework was developed to classify products into:

- Urgent Reorder Required
- Increase Procurement Planning

- Maintain Normal Procurement
 - Reduce Future Procurement
 - Avoid Procurement / Run Clearance
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Growth Trend Analysis

Growth analysis was performed using SQL window functions.

Month-over-Month (MoM) Growth

Analyzed monthly revenue trends and seasonal fluctuations.

Year-over-Year (YoY) Growth

Compared annual business performance.

Regional YoY Growth

Compared revenue growth across regions to identify high-performing and underperforming markets.

9. Power BI Dashboard Development

Stakeholder-focused dashboards were developed.

Executive Overview Dashboard

Provides high-level business performance monitoring.

Features:

- KPI Cards
 - Revenue Trend Analysis
 - Category Performance
 - Revenue through different time in a day
 - Interactive Filters
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Store Manager Dashboard

Supports operational store-level decision making.

Features:

- Daily Revenue Monitoring
 - Weekday vs Weekend Analysis
 - Category Performance
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Sales & Operational Insights Dashboard

Provides sales trends and operational performance insights.

Features:

- Operational performance KPIs
 - Sales Trend Analysis
 - MOM Growth trend
 - YOY Growth trend
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Regional Sales Director Dashboard

Supports regional business performance evaluation.

Features:

- Regional Revenue Comparison
 - Top Performing Regions
 - Regional YoY Growth Analysis
 - Store Performance Comparison
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Inventory Dashboard

Supports inventory planning and procurement decisions.

Features:

- Stock Movement Analysis
 - Dead Stock Identification
 - Fast Moving Products
 - Stockout Risk Monitoring
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10. Key Findings

- North region generated the highest revenue contribution.

- Weekdays generated higher overall revenue compared to weekends.
- Evening hours generated the highest revenue and order volume.
- East region recorded the largest YoY decline.
- Beauty and Fashion categories consistently performed well.
- Q4 showed strong seasonal sales patterns.
- Several fast-moving products require proactive replenishment.
- Dead stock products were identified for inventory optimization.
- Overall business revenue declined by approximately 1.77% in 2025 compared to 2024.
- The analysis also reveals that some categories achieve high revenue with comparatively lower quantity sold. Suggesting premium pricing and higher average selling value products, particularly with Beauty and Electronic categories.

11. Business Recommendations

Sales Strategy

- Increase promotional activities in underperforming regions.
- Replicate successful sales strategies from high-performing regions.

Inventory Strategy

- Prioritize replenishment of fast-moving products.
- Implement clearance campaigns for dead stock items.
- Improve demand forecasting for seasonal products.

Regional Strategy

- Investigate factors driving East region performance decline.
- Continue investment in high-performing North region stores.

12. Team Contributions

Team Member	Contribution
Utkarsh Dhangar	SQL Development, GitHub Management, Executive Dashboard
Neha Bahrela	Excel Cleaning, Documentation, Store Manager Dashboard
Eishu Tamori	Python EDA, Sales & Operational Insights Dashboard and Inventory Insights Dashboard
Jahanvi Pradhan	Excel Cleaning, Regional Director Dashboard

13. Conclusion

The Omnichannel Retail Sales & Inventory Analytics project successfully transformed raw retail data into actionable business intelligence. Through SQL analytics, inventory monitoring, growth analysis, and interactive Power BI dashboards, the project provides valuable insights for operational and strategic decision-making. The solution demonstrates how modern analytics tools can support retail organizations in improving sales performance, inventory efficiency, and overall business growth.